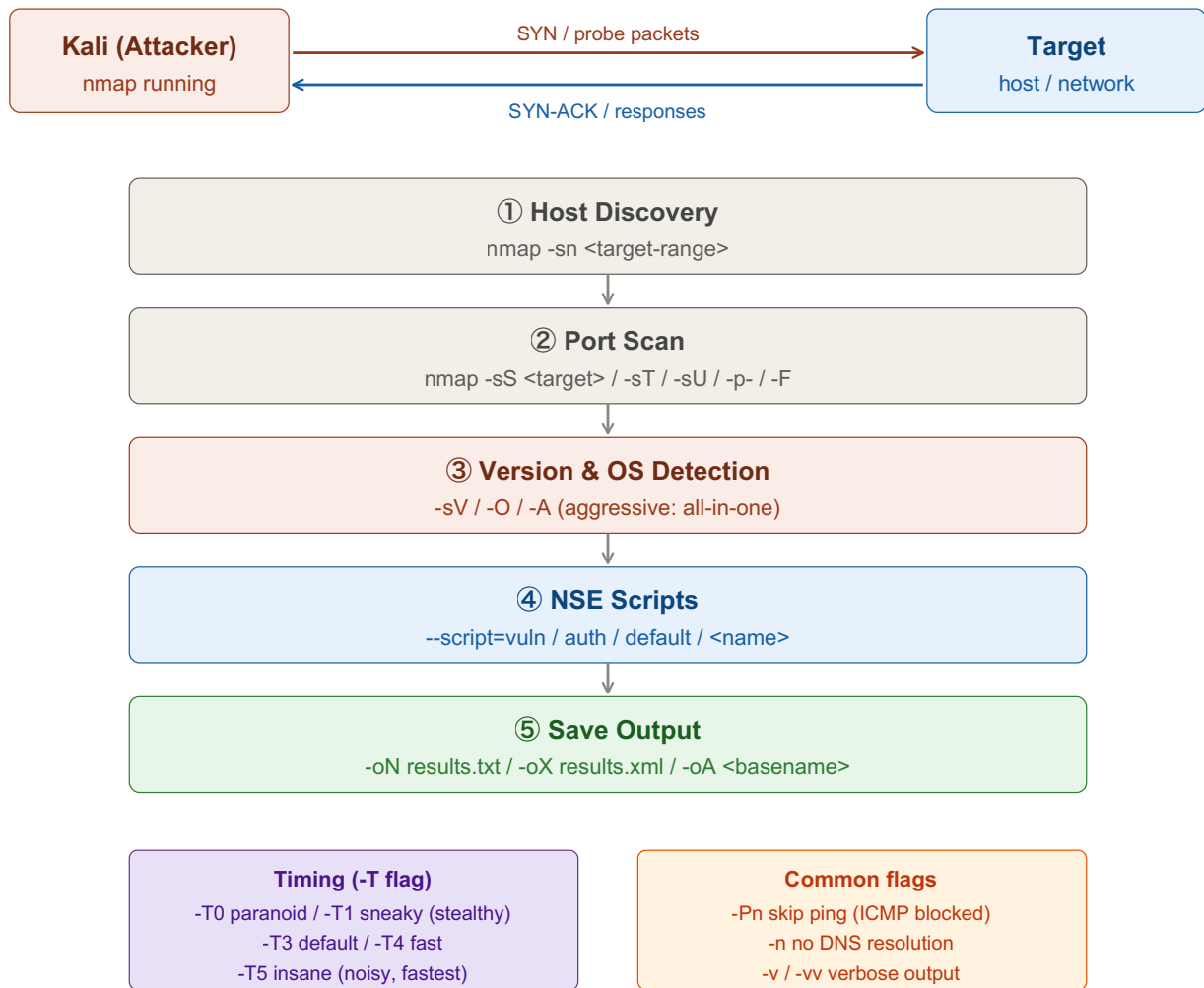


Nmap

Network scanning workflow — host discovery through vulnerability detection. Replace <placeholders> with your own values.



① Host Discovery

Find live hosts before deep-scanning. Use this first so you don't scan dead IPs.

Tab 1 · Kali

```
# Ping sweep — find live hosts
nmap -sn <target-range>

# Examples:
nmap -sn 192.168.1.0/24      # entire subnet
nmap -sn 192.168.1.1-50     # IP range
nmap -sn 192.168.1.5        # single host
```

-Pn skips the ping check and treats all hosts as up — use when ICMP is blocked by a firewall.

② Port Scan

Choose a scan type based on your situation:

```
nmap -sS <target>  # stealth SYN — recommended, requires root
nmap -sT <target>  # TCP connect — no root needed, more noise
nmap -sU <target>  # UDP — slow, use with --top-ports
```

Port selection:

```
-p 22,80,445      specific ports
-p 1-1000          port range
-p-               all 65,535 ports
-F                top 100 ports only
--top-ports <n>   top n most common ports
```

Start with -F or --top-ports 1000 to be quick, then -p- for thoroughness.

③ Version & OS Detection

```
nmap -sV <target>  # service + version detection
nmap -O <target>   # OS fingerprinting (requires root)
nmap -A <target>   # aggressive: -sV + -O + scripts + traceroute
```

-A is great for labs and CTFs. Add -n to skip reverse DNS and speed things up. Use -v or -vv for more detail.

④ NSE Scripts

Run built-in scripts by category or by name:

```
nmap --script=default <target>    # same as -sC
nmap --script=vuln    <target>    # known CVE checks
nmap --script=auth    <target>    # default/weak credential tests
```

Useful individual scripts:

Script	What it does
smb-vuln-ms17-010	EternalBlue (WannaCry) check on port 445
smb-enum-shares	List accessible SMB shares
smb-enum-users	Enumerate users via SMB
ldap-search	Query Active Directory over LDAP (port 389)
ftp-anon	Check for anonymous FTP login
http-title	Grab HTTP page titles from web servers
ssh-brute	Brute-force SSH with a user/pass list

⑤ Save Output

Always save results — you'll need them for reporting or piping into other tools (Metasploit, etc.):

```
nmap -oN results.txt    # human-readable
nmap -oX results.xml    # XML — importable into Metasploit / Nessus
nmap -oA <basename>    # save all three formats at once
```

Common Scan Combos

```
# Quick subnet sweep
nmap -sn 192.168.1.0/24

# Stealth scan with version detection, save output
nmap -sS -sV -T4 -oN results.txt <target>

# Full aggressive scan (all ports)
nmap -A -p- -T4 <target>

# EternalBlue check
nmap -p 445 --script smb-vuln-ms17-010 <target>

# Full vulnerability scan
nmap -sV --script vuln -p- <target>

# UDP top-20 ports
nmap -sU --top-ports 20 <target>
```

Key idea: always start broad (host discovery, top ports) and then narrow down. -A is great for CTFs and labs; use -sS with careful timing for stealthier engagements. Save output with -oA from the start — you don't want to re-run a long scan.